Education

Bachelor of Technology – Instrumentation & Control Engineering
2022 – 2026
CGPA: 6.6

Senior Secondary Education (Class XII)
Percentage: 85.5%
2021

Secondary Education (Class X)
Percentage: 88.8%
2019

Internship Experience

Research Intern

June 2025 – Aug 2025

- Designed scalable **data ingestion pipelines** processing 12,000+ structured and semi-structured system logs.
- Introduced schema normalization and anomaly-filtering layers improving downstream signal quality for AI diagnostics by **~28%**.
- Contributed to an **AI-powered internal knowledge support system** by structuring documentation hierarchies and optimizing retrieval workflows.
- Built performance monitoring frameworks to analyze resolution patterns and system workflow latency.
- Collaborated with engineering teams to formalize reproducible **data contracts**, improving reliability of ML-driven insights.

Analyst Intern

Jan 2025 – Mar 2025

- Redesigned ingestion and validation pipelines for **forensic datasets**, improving data consistency and traceability by **~22%**.
- Strengthened schema governance and multi-stage validation layers to improve data integrity in audit-sensitive environments.
- Conducted workflow dependency analysis and implemented structured modeling improvements for cleaner feature engineering pipelines.

Projects

Context-Aware Document Intelligence Platform (RAG System)

- Built a **Retrieval-Augmented Generation system** enabling contextual question answering over large documents.
- Implemented document parsing, semantic chunking, embedding generation, and vector indexing pipelines.
- Added session-aware memory and structured prompt engineering to improve conversational continuity.

- Designed a modular backend supporting scalable document ingestion workflows.

Multimodal Breast Cancer Risk Assessment System

- Developed a multimodal AI system combining **vision transformer-based mammogram analysis with clinical risk modeling**.
- Integrated imaging features with tabular medical data for improved diagnostic stratification.
- Built an **LLM-powered oncology assistant** with a curated knowledge base for grounded responses.
- Deployed a full-stack platform with scalable API-based inference and persistent patient state management.

Fake News Detection System

- Built a fake news classification system using a **fine-tuned transformer model**.
- Implemented a high-performance inference backend exposing REST APIs for real-time predictions.
- Created a custom pipeline for tokenization, GPU-accelerated inference, and probability-based prediction outputs.

Skills

Programming Languages
Python, JavaScript, SQL

Machine Learning & AI
PyTorch, TensorFlow, CNNs, Vision Transformers, Object Detection, Image Classification, Model Fine-tuning

LLM & GenAI
Large Language Models, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, Agentic Workflows

Web & Full-Stack Development
React, Node.js, Express, MongoDB, Next.js

Other Tools & Technologies
OpenCV, TensorFlow Lite

Soft Skills

Critical Thinking
Team Collaboration
Structured Problem Solving
Time Management
Adaptability